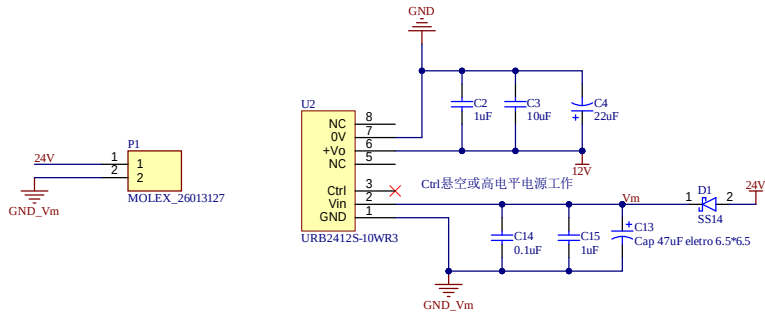
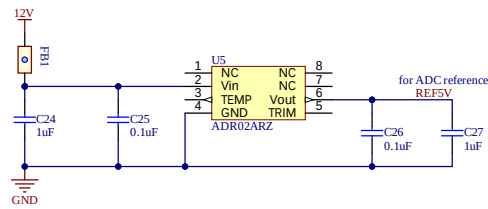
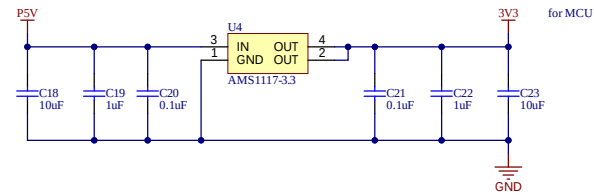


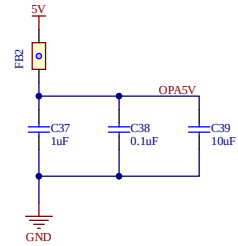
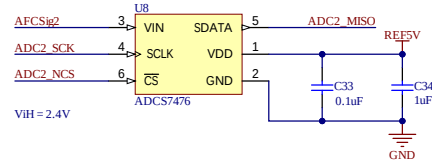
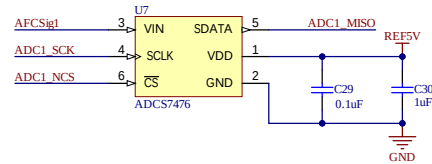
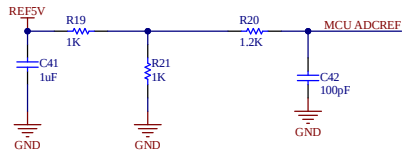
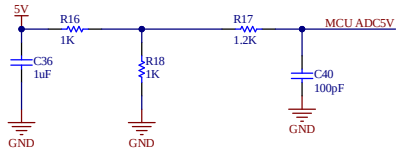
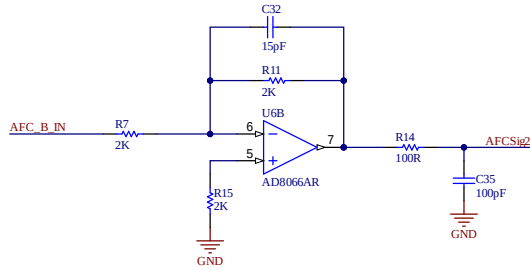
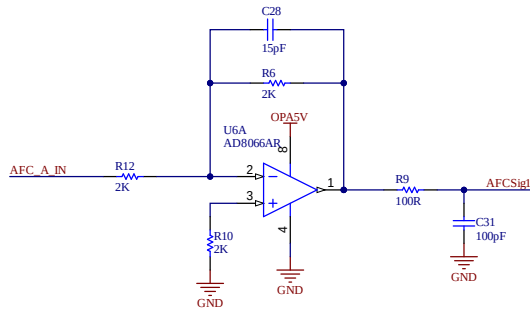
According to datasheet,X7R cap is needed.



更改项1: 添加隔离电源24-12V,去掉7805
更改项2: 更改OPAmpp供电方式

版本 REV
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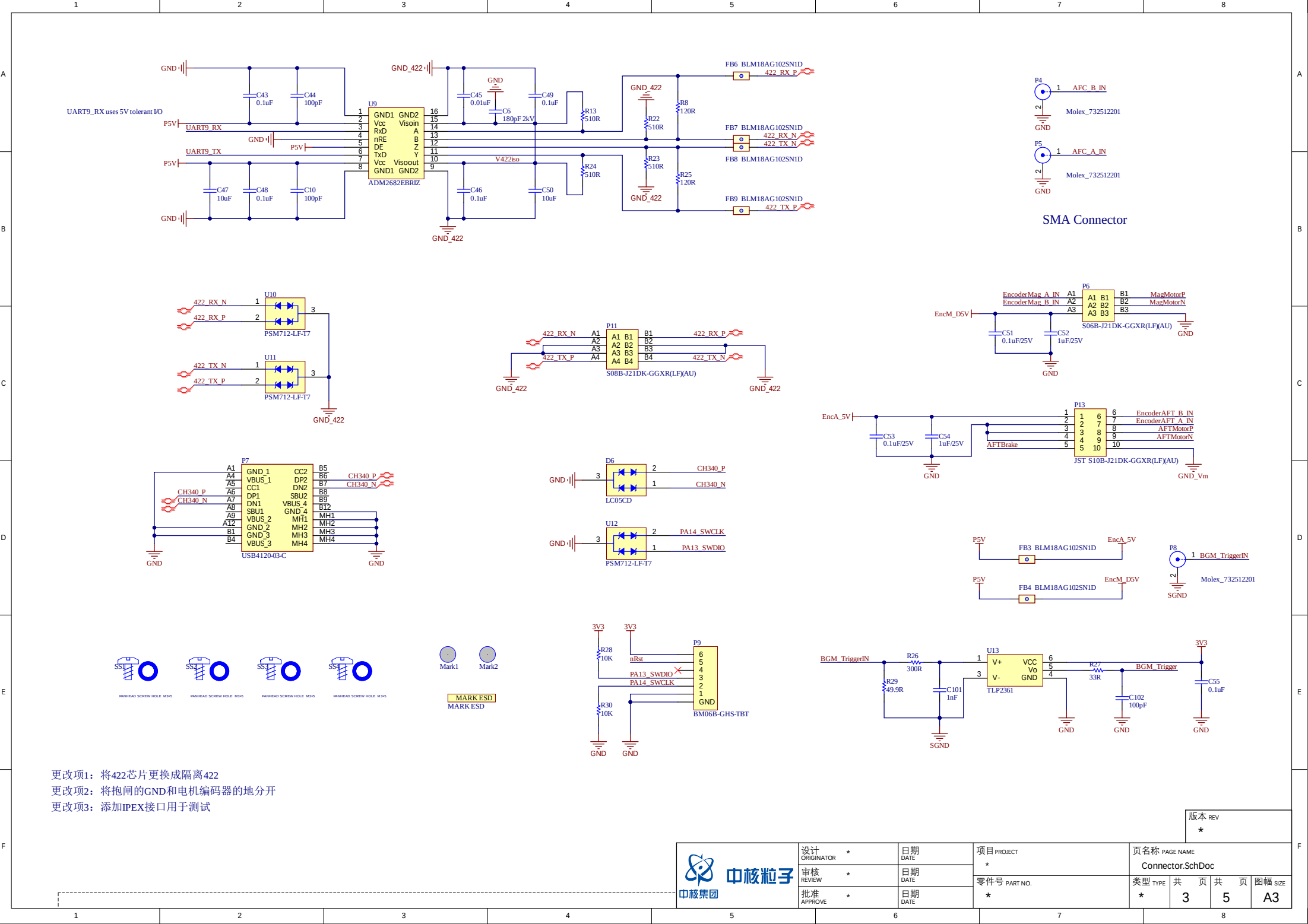
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	审核 REVIEW	*	日期 DATE	*	Power.SchDoc
	批准 APPROVE	*	日期 DATE	零件号 PART NO.	类型 TYPE
				*	共 1 页 共 5 页 图幅 SIZE A3



更改项1: 去掉MCU ADC监测12V的电路（意义不大）
更改项2: adc前级重新设计，更改为反相放大器，添加2K端接（AFT规格书要求）
更改项3: 添加IPEX焊盘，可以用一根BNC转IPCX同轴线直接测量板端analog信号

版本 REV
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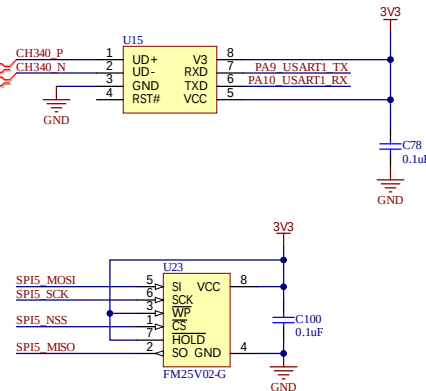
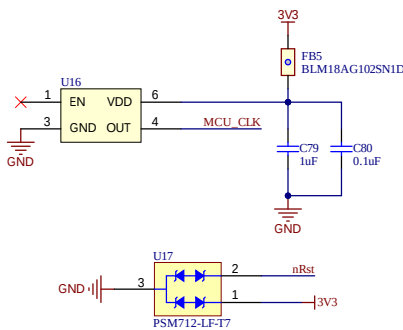
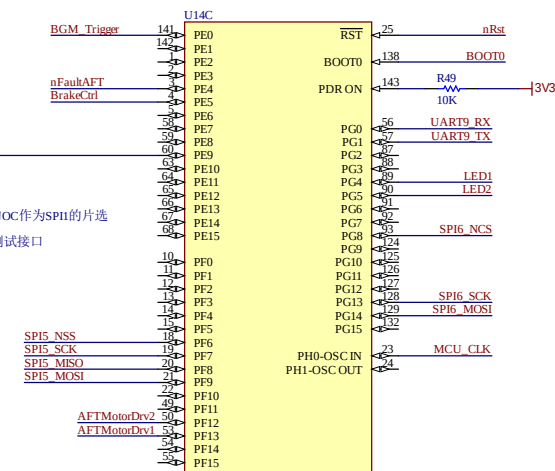
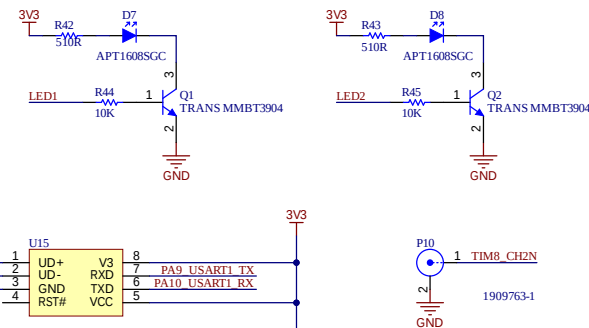
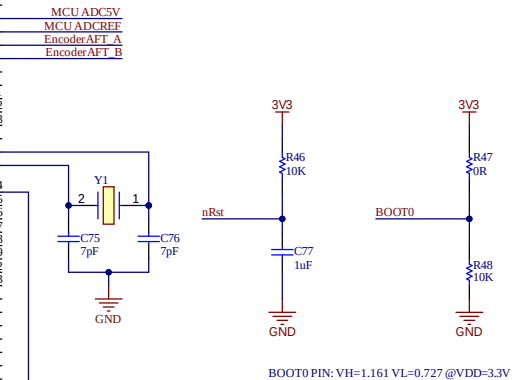
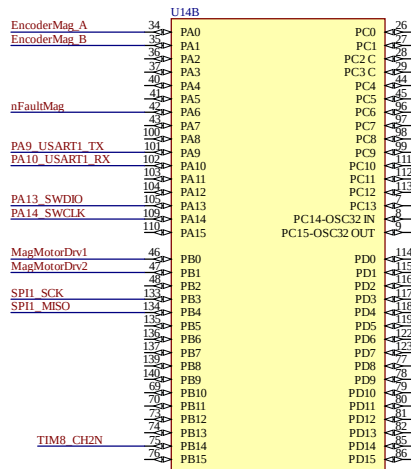
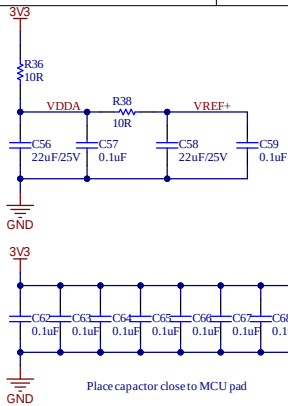
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	批准 APPROVE	*	日期 DATE	零件号 PART NO.	类型 TYPE	共 页	共 页	图幅 SIZE
				*	*	2	5	A3



更改项1: 将422芯片更换成隔离422
更改项2: 将抱闸的GND和电机编码器的地分开
更改项3: 添加IPEX接口用于测试

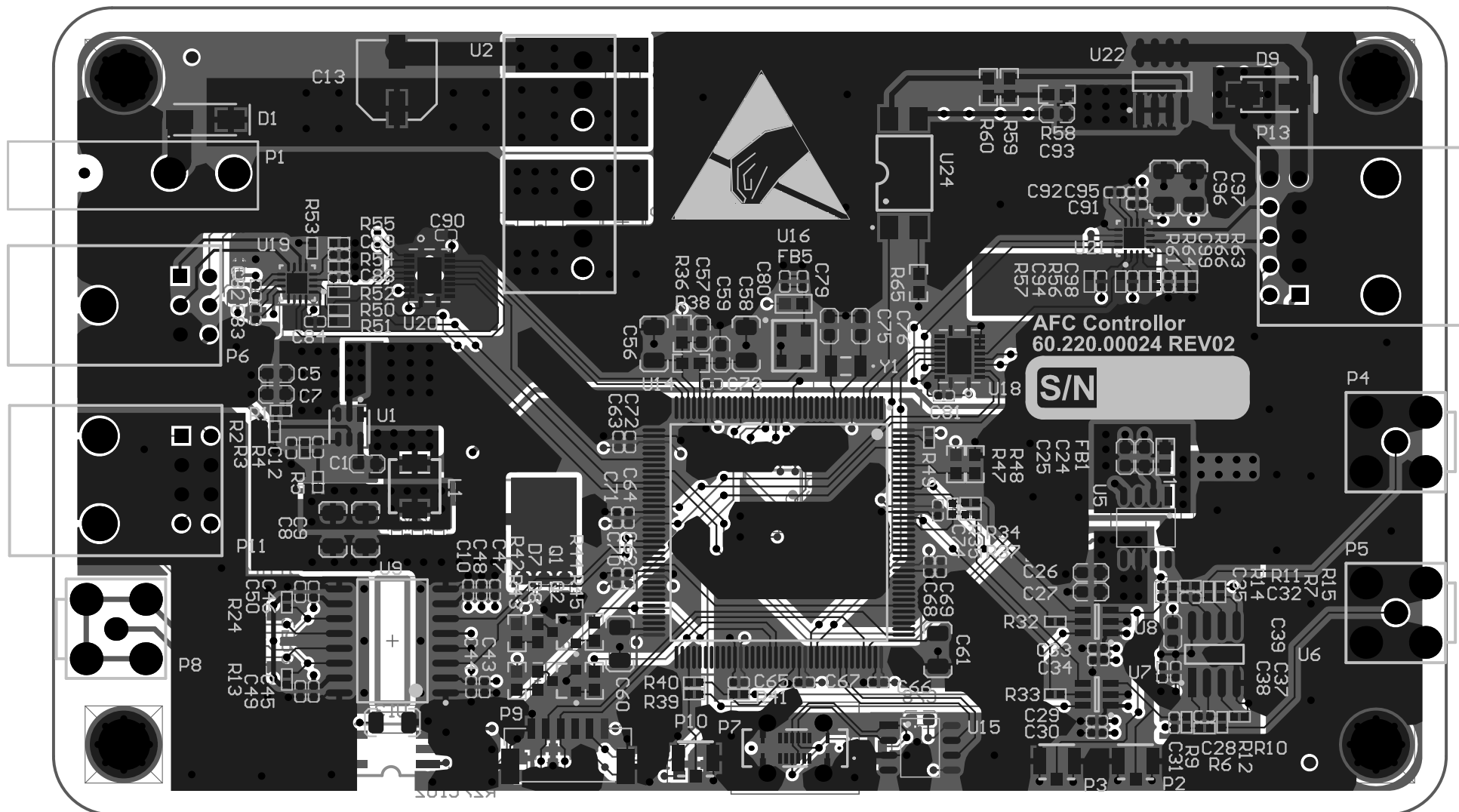
版本	REV
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 中核集团	设计 ORIGINATOR	*	日期 DATE	项目 PROJECT	页名称 PAGE NAME
	审核 REVIEW	*	日期 DATE	*	Connector.SchDoc
批准 APPROVE	*	日期 DATE		零件号 PART NO.	类型 TYPE
				*	
					共 3 页
					共 5 页
					图幅 SIZE
					A3



更改项4: 删掉两个LED

版本 REV
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Comment	Description	Designator	Footprint	Label	Quantity
13uF	ceramic 50V, ceramic 10V, ceramic 25V	C1, C7, C14, C20, C21, C25, C26, C35, C37, C38, C78, C93	C0603(1608)	Cap 0.1uF 0603-50V, Cap 0.1uF 0603-10V, Cap 0.1uF 0603	12
10uF	ceramic 25V	C2	C0805(2012)	Cap 1uF 0805-25V	3
10uF	ceramic 25V	C3	C0805(2012)	Cap 10uF 0805-25V	1
22uF	TANT 25V	C4	C-31126032	Cap 22uF 6032-25V	1
100pF	ceramic 25V	C5, C18, C23, C39	C0603(1608)	Cap 10pF 0603-25V	4
100pF	ceramic 25V	C9	C1206(3216)	Cap 150pF 1206-25V	1
22uF/25V	1206 (3216 080) 22 uF, 25 V, ± 10%, X5R, ±0.1% B	C8, C9, C36, C38, C60, C81	C1206(3216)	Cap 22uF 1206-25V	6
100pF	Ceramic 50V, ceramic 10V	C10, C31, C33, C40, C42, C44	C0402(1005)	Cap 100pF 0402	6
100pF	ceramic 50V	C12	C0402(1005)	Cap 75pF 0402-50V	1
Cap 4.7uF electro 4.5*6.5	electro 50V Panasonic KEET1H470AP 4.5*6.5	C13	CAPE 6x6 6mm	Cap 47uF electro 4.5*6.5	1
1uF	ceramic 50V	C15, C19, C22, C24, C27	C0603(1608)	Cap 1uF 0603-50V	4
2.2uF	ceramic 25V	C16, C17	C0805(2012)	Cap 2.2uF 0805-25V	2
15pF	ceramic 25V	C28, C32	C0402(1005)	Cap 15pF 0402	2
		C29, C33, C34, C41, C46, C48, C49, C62, C63, C64, C65, C66, C67, C68, C70, C71, C72, C73, C74, C86, C81, C82, C84, C85, C88, C90, C91, C94, C95, C96, C100	C0402(1005)	Cap 0.1uF 0402-25V, Cap 0.1uF 0402-10V, Cap 0.1uF 0402-50V	32
1uF	ceramic 10V	C30, C34, C36, C37, C43, C77, C79	C0402(1005)	Cap 1uF 0402-10V	7
0.01uF	ceramic 25V	C45	C0402(1005)	Cap 0.01uF 0402-25V	1
10uF	ceramic 10V	C47, C50	C0402(1005)	Cap 10uF 0402	2
0.1uF/25V	ceramic 25V	C51, C59	C0603(1608)	Cap 0.1uF 0603-25V	2
1uF/25V	ceramic 25V	C52, C54	C0603(1608)	Cap 1uF 0603-25V	2
ceramic 50V COG	ceramic 50V COG	C75, C76	C0603(1608)	Cap 7pF 0603-50V 250	2
22pF	ceramic 25V	C83, C92	C0402(1005)	Cap 22pF 0402-25V	2
0.1uF/35V	ceramic 35V	C86, C96	C1206(3216)	Cap 50 uF 1206 (3216)	2
415pF	MLCC 47uF 25V	C87, C87	C1206(3216)	Cap 47uF 1206-25V	2
470pF	ceramic 50V	C93, C99	C0402(1005)	Cap 470pF 0402-50V	1
100pF	ceramic 10V	C101	C0603(1608)	Cap 1uF 0603-10V	1
100pF	ceramic 25V	C102	C0603(1608)	Cap 100pF 0603-25V	1
SS14	Schottky diode	D1	PCNAMS1675242N	SS14	1
TVS FOR USB2.0	TVS FOR USB2.0	D6	SO723-3	LE05C0	1
401160855C	MINILED green 0605	D8	LE060502	LED 401160855C	1
Hi-Zener 二极管 20V 2.1A	Hi-Zener 二极管 20V 2.1A	D9	SMA-2_B_V	SS18	1
BLM18AG102SN1D	FERRITE BEAD 1.0KHz 100MHz	F61, F62, F63, F64, F65, F66, F67, F68, F69	F8003(1608)	FB 3K@100MHz 0603	9
FXL0420-100-M	10uH 1.6A 2*4.4*1.8mm	L1	FXL0420-69B-M	FXL0420-100-M	1
Mark	Mark	Mark1, Mark2	Mark	Mark	2
MARK ESD	MARK ESD	Mark ESD	Mark ESD	MARK ESD	1
MAX98130A327	U-connector	P2	06031127	06031127	1
UG29763-1	PEX PCB	P2, P3, P10	1509763-1	1509763-1	3
Molex_72512201	Molex_72512201 SMA Right angle	P4, P5, P8	Molex_72512201	Molex_72512201	3
S08B-21DK-SS08B (9AU)	H6	S08B-21DK-GGVR	S08B-21DK-GGVR(9AU)	S08B-21DK-SS08B (9AU)	1
USB Connectors USB C Rec 3u" vert 16P MAT & TH stakes prep T+H+C+H = 2.28	P7	USB412003C	USB4120-03-C	USB4120-03-C	1
S08B-04HS-TBT	P9	S08B-04HS-TBT	S08B-04HS-TBT	S08B-04HS-TBT	1
S08B-21DK-SS08B (9AU)	P11	S08B-21DK-GGVR	S08B-21DK-SS08B (9AU)	S08B-21DK-SS08B (9AU)	1
BT S10B-21DK-SS08B (9AU)	P13	BT S10B-21DK-GGVR	BT S10B-21DK-SS08B (9AU)	BT S10B-21DK-SS08B (9AU)	1
TRANS LMH7292A	Q1, Q2	Q1, Q2	MMBT1904	MMBT1904	1
R11K	R1	R0402(1005)	Res 11K 0402	1	
R10K	R2	R0402(1005)	Res 10K 0402	1	
R13K	R3	R0402(1005)	Res 13.3K 0402	1	
R10K	R4	R0402(1005)	Res 10K 0402	1	
R49.9K	R5	R0402(1005)	Res 49.9K 0402	1	
2K	R6, R7, R10, R11, R12, R13	R0402(1005)	Res 2K 0402	6	
120R	Res 120R 0402	R6, R25	R0402(1005)	2	
100R	Res 100R	R9, R14	R0402(1005)	2	
R10R	Res 10R	R12, R22, R23, R24	R0402(1005)	4	
4K	Res 4K	R16, R18, R19, R21	R0402(1005)	4	
1.2K	Res 1.2K	R17, R20	R0402(1005)	2	
R20R	Res 20R	R22	R0603(1608)	1	
R39	Res 39R	R27	R0603(1608)	1	
10K	Res 10K	R29, R30, R44, R45, R46, R58	R0603(1608)	7	
49.9R	Res 49.9R	R28	R0603(1608)	1	
R39	Res 39R	R32, R33, R34, R37, R39, R41	R0402(1005)	6	
R6	Res 6R	R35, R40	R0402(1005)	2	
10R	Res 10R	R36, R38	R0603(1608)	2	
S10R	Res 10R	R42, R43	R0603(1608)	2	
R6	Res 6R	R47	R0603(1608)	1	
10K	Res 10K	R49, R50, R51, R56, R57	R0402(1005)	6	
1.3K	Res 1.3K	R52, R53	R0402(1005)	2	
20K	Res 20K	R53, R55	R0402(1005)	2	
5.1K	Res 5.1K	R54, R56	R0402(1005)	2	
R6	Res 6R	R55	R0402(1005)	1	
4.7K	Res 4.7K 0603 1%	R59	R0603(1608)	1	
12.1K	Res 12.1K	R60	R0603(1608)	1	
100R	Res 100R	R65	R0603(1608)	1	
4.7K	Res 4.7K	R66	R0603(1608)	1	
PAN-HEAD SCREW HOLE M3*5	SS1, SS2, SS3, SS4	MCN/NTING_HOLE_3	Power Standoff hole_3mm	4	
FP554302DCR	IC REG BUCK ADJ 3A 0.0722A	U1	FP-DC0300A-IPC	FP554302DCR	1
VRB24125-10WR3	Isolated DCDC 24V-12V Maximum	U2	VRB24125-10WR3	VRB24125-10WR3	1
ACPT118AU1Z-5.0-87	DO to SV low noise 100mA	U3	TSOT-23-5	ACPT118AU1Z-5.0-87	1
AMS1117-3.3	AMS1117-3.3	U4	SO723-3	AMS1117-3.3	1
ADCS2042	OP AMP VFB 145MHz RRO SCOP	U5	SCOP	ADCS2042	1
AD8066AR	OPAMP VFB 145MHz RRO SCOP	U6	SCOPB_N	AD8066AR	1
ADCS2476	ADC 12BIT 1MSPS SO7-23.6	U7, U8	ADCS2476	ADCS2476	2
ADM2642EBRIZ	2.4V rms Signal and Power Isolated RS-485 Transceiver	U9	SOIC16 Body 10.3x7.5mm	ADM2642EBRIZ	1
PSM712-1F-17	FVS	U10, U11, U22, U27	SO723-3	PSM712-1F-17	6
TLP2361	fast optocoupler invert Logic Toshiba	U13	SOS	TLP2361	1
IC MCU S28T1 1MB FLASH 144QFP	U14	LOFP144-1A-IPC_A	STM32H732076	IC MCU S28T1 1MB FLASH 144QFP	1
LSB87TL181L	U15	SOIC8	C1436N	LSB87TL181L	1
DC/DC XO 25.000MHz IC MOS SMD	U16	ICS-3225533-250-Pin 39	ICS-3225533-250-Pin 39	DC/DC XO 25.000MHz IC MOS SMD	1
SN74VLC245ARGYR	Octal Bus Transceiver with 3-State Outputs	U18, U20	SN74VLC245ARGYR	SN74VLC245ARGYR	2
Motor Motion/ignition Controllers & Drivers Automotive adjustable 1-A low-dropout (LDO) linear regulator with power good 6-VSON -40 to 125	U19, U21	PP-AGT0018C-IPC_C	DRV8987BRGTR	Motor Motion/ignition Controllers & Drivers Automotive adjustable 1-A low-dropout (LDO) linear regulator with power good 6-VSON -40 to 125	2
RP9917	MOSFET, P沟道, -20V, 18A, SOIC	U22	SOIC8_N	RP9917	1
FM25V02-G	FM25V02-G	U23	FM25V02-G SCOPB	FM25V02-G	1
PC817C2C39P	photocoupler sharp buffer logic	U24	SMD-4P	PC817C2C39P	1
IC352-1P-20PMM	U1	IC-3257P20PMM	IC352-1P-20PMM	IC352-1P-20PMM	1